

POKROVSK, Russian Federation

WMO: 248560

Lat: 61.483N

Lon: 129.150E

Elev: 121

StdP: 99.88

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -50.7	(c) -48.9	(d) -52.1	(e) 0.0	(f) -50.3	(g) -50.4	(h) 0.0	(i) -49.0	(j) 6.1	(k) -35.5	(l) 5.4	(m) -35.0	(n) 1.0	(o) 240	(p) 0.726

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.3	30.1	18.8	27.9	18.1	25.8	17.2	20.4	27.1	19.3	25.7	18.2	24.1	2.8	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.2	13.3	23.2	16.9	12.2	22.3	15.7	11.3	21.3	59.4	27.1	55.8	25.9	52.1	24.3	27.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 6.7	(b) 5.9	(c) 5.3	(e) -52.3	(f) 33.3	(g) 2.4	(h) 1.6	(i) -54.0	(j) 34.4	(k) -55.4	(l) 35.3	(m) -56.7	(n) 36.2	(o) -58.5	(p) 37.3		
(a) 6.7	(b) 5.9	(c) 5.3	(e) -52.3	(f) 22.5	(g) 2.4	(h) 1.8	(i) -54.0	(j) 23.8	(k) -55.4	(l) 24.8	(m) -56.7	(n) 25.8	(o) -58.5	(p) 27.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-9.0	-38.3	-33.8	-20.7	-5.0	6.8	15.2	18.3	14.5	5.4	-7.5	-26.8	-37.9		
	DBStd	21.52	6.33	6.53	7.62	6.19	4.23	4.12	3.79	3.75	4.22	6.67	8.72	6.70		
	HDD10.0	7538	1498	1228	952	452	118	6	1	6	146	542	1104	1486		
	HDD18.3	10061	1757	1461	1210	702	359	109	48	128	387	801	1354	1744		
	CDD10.0	594	0	0	0	0	17	163	259	146	9	0	0	0		
	CDD18.3	74	0	0	0	0	0	16	48	11	0	0	0	0		
	CDH23.3	1067	0	0	0	0	8	293	612	152	2	0	0	0		
	CDH26.7	313	0	0	0	0	1	78	200	34	0	0	0	0		
(14) Wind	WSAvg	2.4	1.8	2.0	2.4	2.7	3.0	2.7	2.5	2.6	2.8	2.5	2.1	1.8		
(15) Precipitation	PrecAvg	273	10	7	6	8	24	35	51	48	34	22	15	11		
	PrecMax	394	28	13	15	28	73	87	115	141	94	47	39	30		
	PrecMin	128	4	3	1	1	3	10	13	9	12	5	5	4		
	PrecStd	62	5	3	4	6	17	20	31	29	21	10	9	5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-18.5	-9.3	2.9	13.3	24.3	31.5	33.4	30.1	21.3	7.3	-0.9	-15.2		
		MCWB	-18.9	-10.1	-0.5	6.0	12.8	18.5	20.2	19.3	13.2	3.9	-1.6	-15.6		
	2%	DB	-22.7	-16.4	-1.1	9.2	20.5	28.9	31.0	27.0	17.6	4.7	-5.3	-21.5		
		MCWB	-22.9	-16.8	-3.9	3.8	10.7	17.4	19.6	18.3	11.4	2.1	-6.0	-21.8		
	5%	DB	-25.7	-20.6	-4.3	6.6	18.0	26.7	29.0	24.7	15.2	2.7	-10.1	-25.2		
		MCWB	-25.9	-20.9	-6.1	1.9	9.3	16.5	18.8	16.9	10.5	0.8	-10.6	-25.4		
	10%	DB	-28.9	-23.9	-7.8	4.4	15.6	24.5	26.9	22.6	13.1	1.0	-14.1	-28.3		
		MCWB	-29.0	-24.1	-9.1	0.7	8.3	15.6	18.2	16.0	9.5	-0.5	-14.5	-28.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-18.8	-10.1	-0.6	6.2	13.5	20.4	22.1	21.2	14.2	4.8	-1.3	-15.5		
		MCDB	-18.5	-9.4	2.5	12.6	22.8	28.0	29.3	27.0	18.9	6.7	-1.0	-15.2		
	2%	WB	-22.9	-16.8	-3.6	4.0	11.4	18.6	21.0	19.4	12.5	2.5	-6.0	-21.7		
		MCDB	-22.7	-16.5	-1.3	8.5	18.6	26.3	27.9	24.9	16.1	4.1	-5.2	-21.5		
	5%	WB	-25.9	-20.9	-6.1	2.3	10.1	17.4	20.0	18.0	11.1	0.9	-10.6	-25.4		
		MCDB	-25.7	-20.7	-4.4	5.9	16.5	24.7	26.8	23.4	14.2	2.6	-10.2	-25.3		
	10%	WB	-29.0	-24.1	-9.1	0.9	8.8	16.2	19.1	16.6	9.7	-0.5	-14.5	-28.4		
		MCDB	-28.9	-23.9	-7.8	4.1	14.6	23.2	25.3	21.5	12.8	1.0	-14.2	-28.3		
(35) Mean Daily Temperature Range	MDBR		7.7	11.0	15.7	13.8	11.6	13.2	12.3	12.3	10.5	8.4	9.0	7.0		
	5% DB	MCDBR	11.5	13.5	16.7	13.9	16.9	17.5	16.5	16.7	15.2	9.1	11.0	10.5		
		MCWBR	11.4	13.2	14.6	9.1	9.0	7.9	7.2	8.7	9.6	7.0	10.5	10.5		
	5% WB	MCDBR	11.6	13.4	16.7	12.7	14.6	15.2	13.5	13.6	12.5	8.1	10.8	10.5		
		MCWBR	11.4	13.2	14.6	8.6	8.1	7.7	6.6	7.6	8.4	6.5	10.5	10.4		
(40) Clear-Sky Solar Irradiance	taub		0.223	0.264	0.305	0.365	0.434	0.429	0.515	0.488	0.370	0.292	0.229	0.190		
	taud		1.973	2.066	2.061	2.065	2.096	2.224	1.964	1.995	2.289	2.300	2.005	1.887		
	Ebn at Noon		474	698	794	812	781	797	702	682	725	671	452	327		
	Edh at Noon		41	74	107	132	140	126	160	143	88	59	38	25		
(44) All-Sky Solar Radiation	RadAvg		0.40	1.42	3.18	5.07	5.23	5.92	5.42	4.21	2.60	1.19	0.54	0.20		
	RadStd		0.02	0.05	0.11	0.20	0.39	0.40	0.34	0.29	0.27	0.13	0.04	0.01		

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.56	N/A	N/A	N/A	N/A	N/A	-205	N/A	N/A	N/A	(46)
(47)	Regional (5 neighbors)	+0.71	+1.43	+1.26	N/A	N/A	N/A	-236	-246	N/A	N/A	(47)

Nomenclature: See separate page